

Example 3 (Bayes rule)

①

Let A = successful well

A' = unsuccessful well

B = detailed test

Given $P(A) = 0.4$, $P(A') = 0.6$, $P(B|A) = 0.6$ and

$$P(B|A') = 0.2$$

(a) Calculate $P(A|B)$.

(b) Construct the contingency table with the probabilities.

(c) Calculate $P(A|B')$.

(d) Construct a tree diagram with the probabilities.

(e) Construct a Venn diagram with the probabilities.

Solutions:

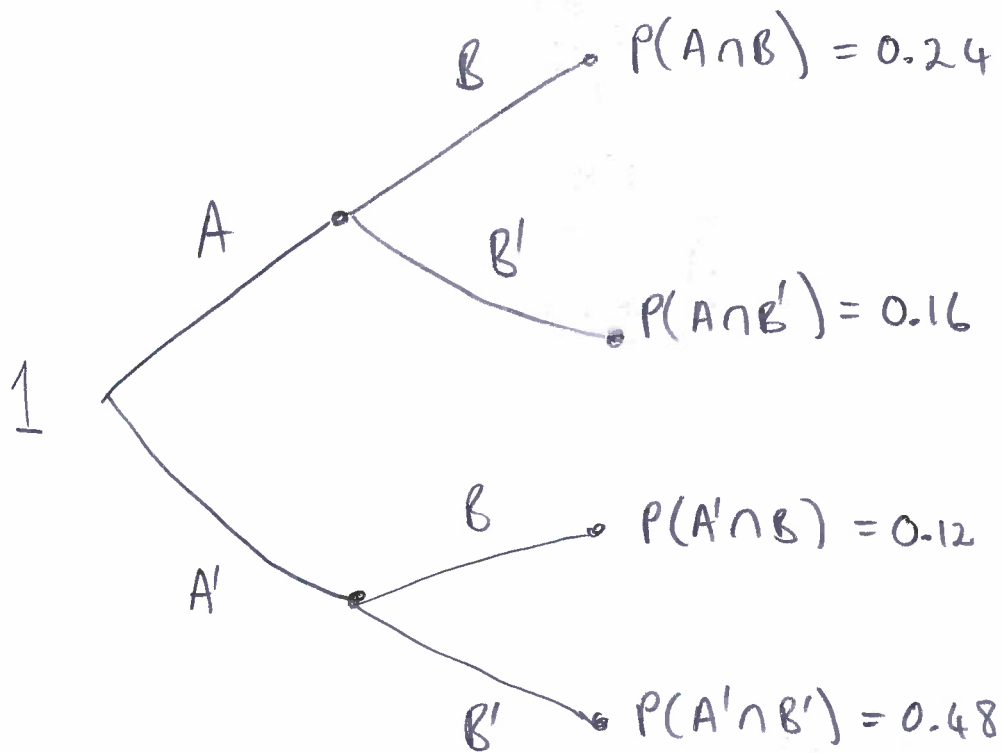
(a)
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$= \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A')P(A')}$$
$$= \frac{(0.6)(0.4)}{(0.6)(0.4) + (0.2)(0.6)}$$
$$= \frac{0.24}{0.36} = \underline{0.6667}$$

(b)

	A	A'	tot
B	$P(A \cap B) = 0.24$	$P(A' \cap B) = 0.12$	$P(B) = 0.36$
B'	$P(A \cap B') = 0.16$	$P(A' \cap B') = 0.48$	$P(B') = 0.64$
tot	$P(A) = 0.4$	$P(A') = 0.6$	1

(c)
$$P(A|B') = \frac{P(A \cap B')}{P(B')}$$
$$= \frac{0.16}{0.64} = \underline{0.25}$$

(d)



Note:

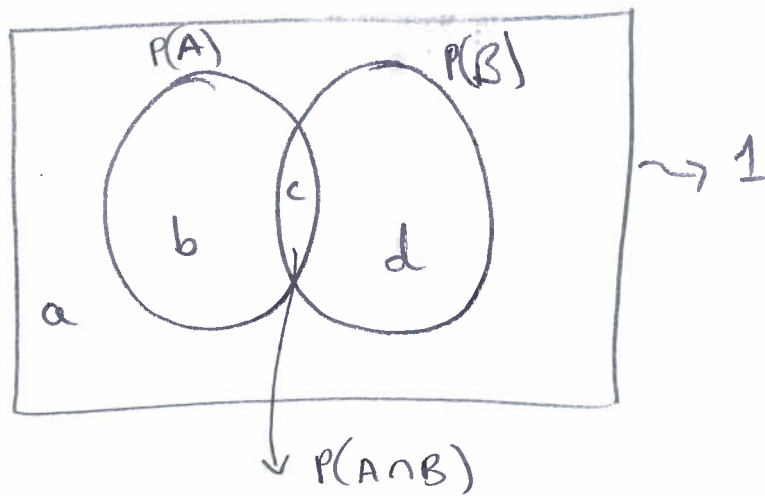
$$P(A) = P(A \cap B) + P(A \cap B') = 0.4$$

$$P(A') = P(A' \cap B) + P(A' \cap B') = 0.6$$

$$P(B) = P(A \cap B) + P(A' \cap B) = 0.36$$

$$P(B') = P(A \cap B') + P(A' \cap B') = 0.64$$

(e)

Note :

$$P(A) = 0.4 \quad \underline{\{b, c\}}$$

$$P(A') = 0.6 \quad \underline{\{a, d\}}$$

$$P(B) = 0.36 \quad \underline{\{c, d\}}$$

$$P(B') = 0.64 \quad \underline{\{a, b\}}$$

$$P(A \cap B) = 0.24 \quad \underline{\{c\}}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.52 \quad \underline{\{b, c, d\}}$$

$$P(A' \cap B') = 1 - P(A \cup B) = 0.48 \quad \underline{\{a\}}$$

$$P(A' \cap B) = P(B) - P(A \cap B) = 0.12 \quad \underline{\{d\}}$$

$$P(A \cap B') = P(A) - P(A \cap B) = 0.16 \quad \underline{\{b\}}$$